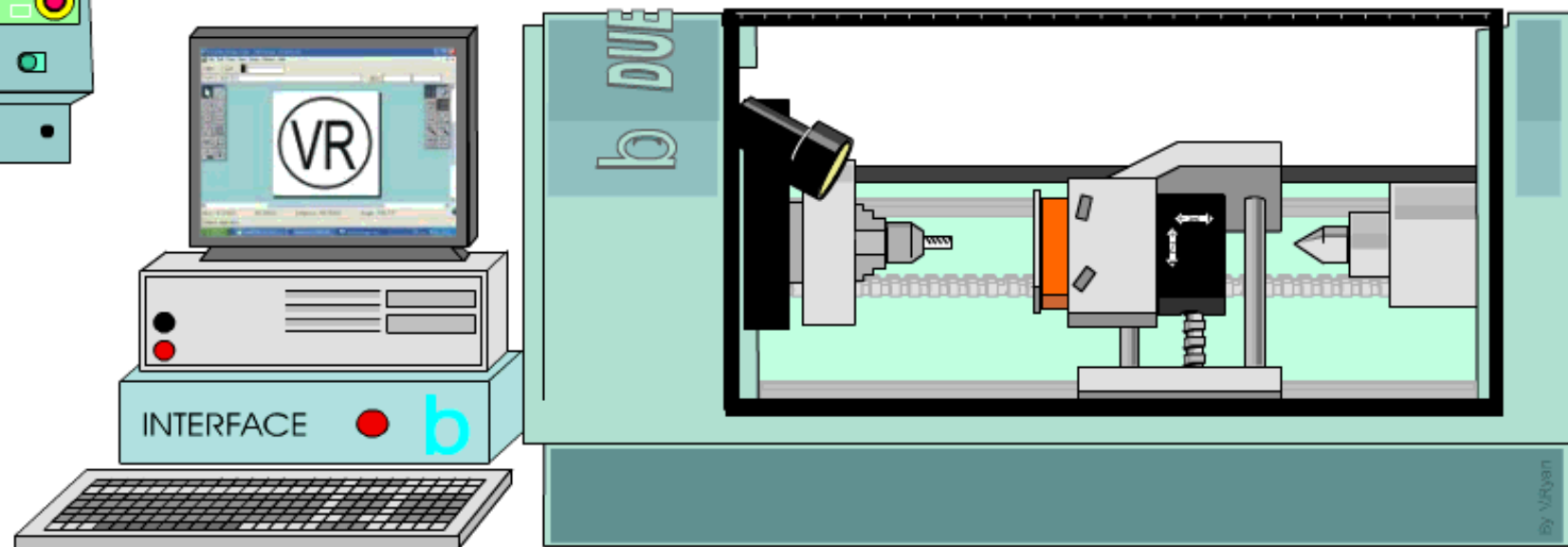
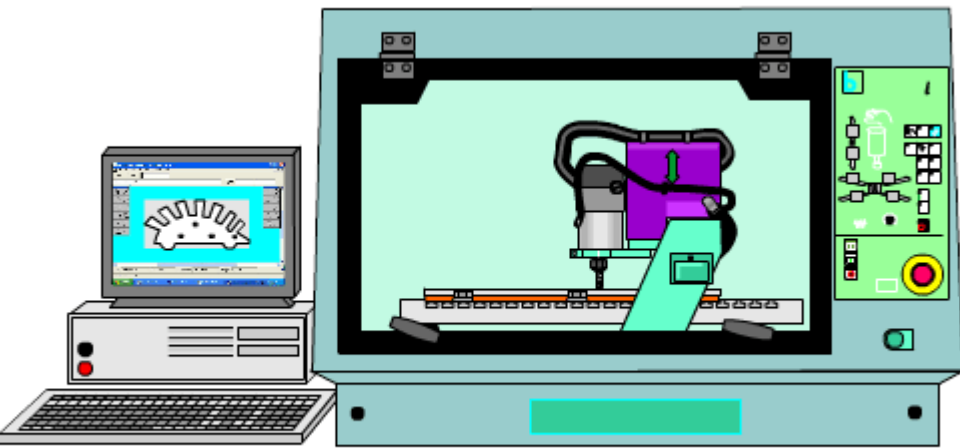


Numerical Control

COMPUTER NUMERICAL CONTROL
C N C



Numerical Control

- a form of programmable automation in which the process is controlled by numbers, letters, and symbols
- a program of instructions designed for a particular work-part or job
- When the job changes, the program of instructions is changed.
- This capability to change the program for each new job is what gives NC its flexibility

BASIC COMPONENTS OF AN NC SYSTEM

1. Program of instructions
2. Controller unit, also called a machine control unit (MCU)
3. Machine tool or other controlled process

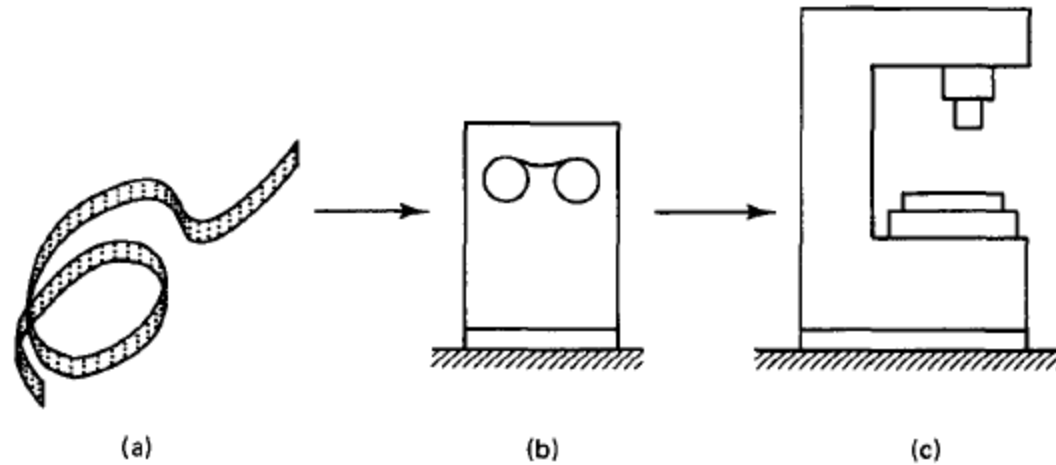


FIGURE 7.1 Three basic components of a numerical control system: (a) program of instruction; (b) controller unit; (c) machine tool.

1. Program of instructions

- The program of instructions is the detailed step-by-step set of directions which tell the machine tool what to do.
- It is coded in numerical or symbolic form on some type of input medium that can be interpreted by the controller unit.
- *The part program is converted into a sequence of machine tool actions by means of the input medium, which contains the program the controller unit, which interprets the input medium.*
- The controller unit and the input medium must be compatible - the input medium uses coded symbols which represent the part program, and the controller unit must be capable of reading those symbols

PUNCHED TAPE IN NC

- There are eight regular columns of holes running in the lengthwise direction of the tape.
- There is also a ninth column of holes between the third and fourth regular columns. However, these are smaller and are used as sprocket holes for feeding the tape.
- The coding of the tape is provided by either the presence or absence of a hole in the various positions.

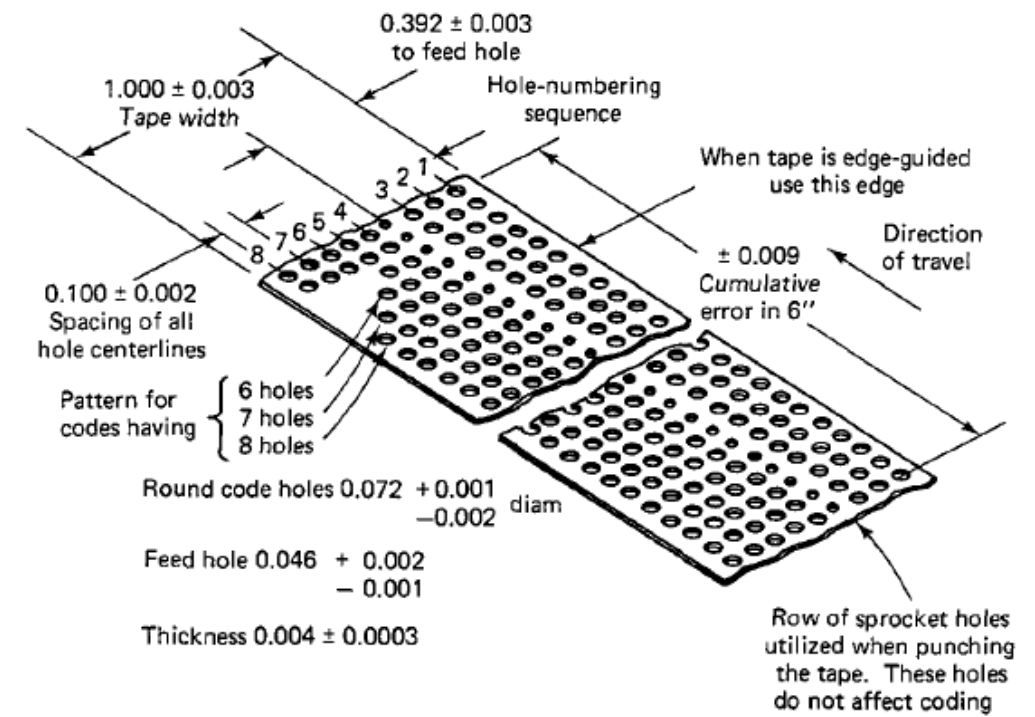
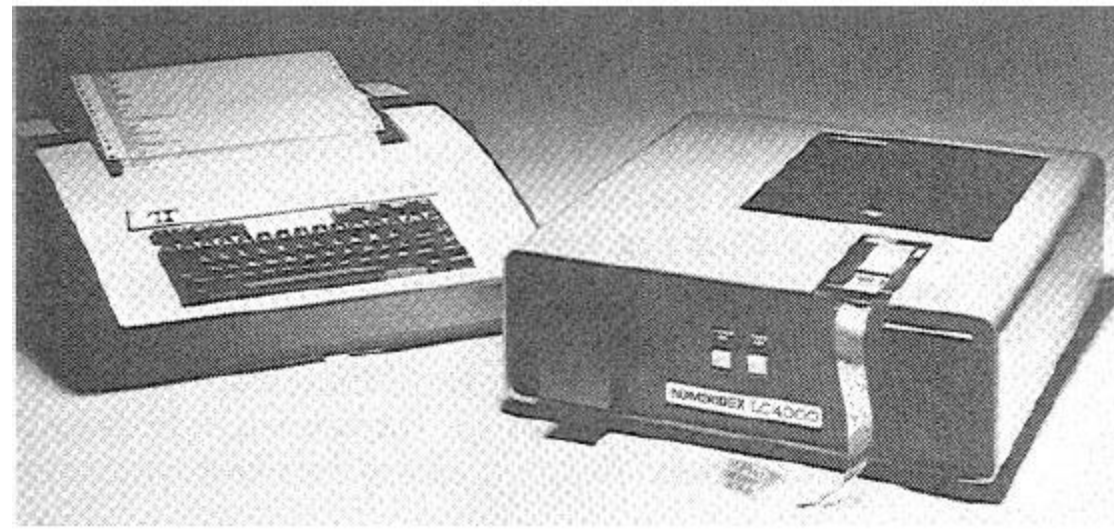


FIGURE 8.1 Numerical control punched tape format as standardized by Electronic Industries Association. (Reprinted from Childs [2].)

- Because there are two possible conditions for each position—either the presence or absence of a hole—this coding system is called the binary code.
- The NC tape coding system is used to code not only numbers, but also alphabetical letters and other symbols.
- Eight columns provide more than enough binary digits to define any of the required symbols.
- A character is a combination of bits, which represents a letter, number, or other symbol.
- A word is a collection of characters used to form part of an instruction.

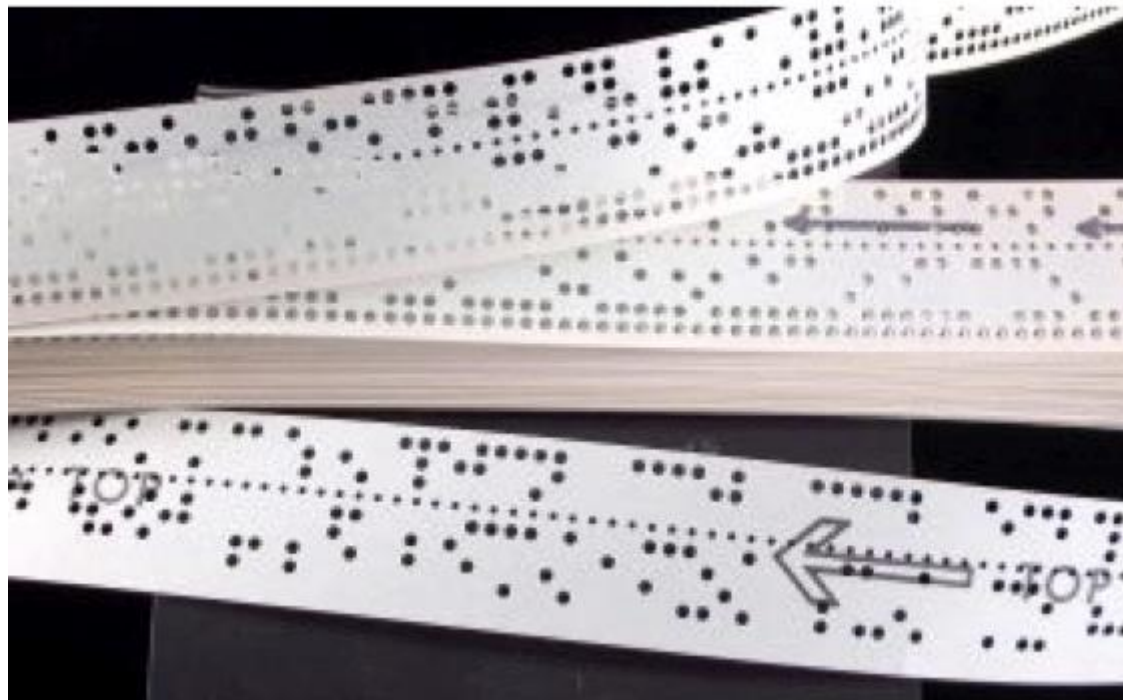
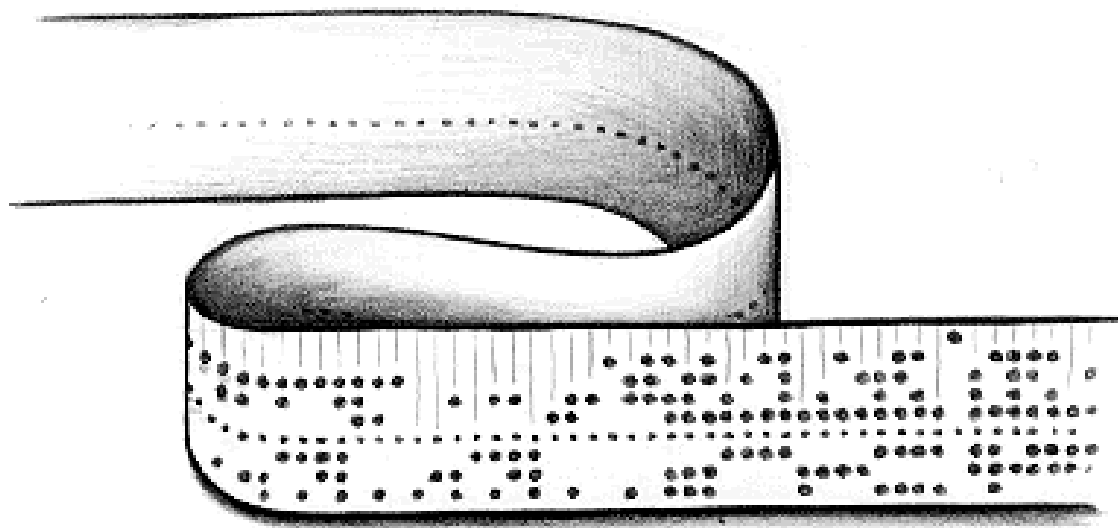
Preparing the punched tape

- There are two basic methods of preparing the punched tape.
- The first method is associated with manual part programming and involves the use of a typewriterlike device.
- After the last instruction has been read into the controller, the tape is rewound back to the start of the program to be ready for the next work-part.



8.2 Equipment used to prepare punched tape for NC. (Courtesy of Numeridex Corp.)

- The second method is used with computer-assisted part programming - by this approach, the tape is prepared directly by the computer using a device called a tape punch.
- During production on a conventional NC machine, the tape is fed through the tape reader once for each workpiece.
- It is advanced through the tape reader one instruction at a time.
- While the machine tool is performing one instruction, the next instruction is being read into the controller unit's data buffer.
- This makes the operation of the NC system more efficient.



Binary Symbol Table

A	□■□□ □□■	a	□■□□ □□■	0	□□■□ □□□□
B	□■□□ □□□	b	□■□□ □□■	1	□□■□ □□□■
C	□■□□ □□■	c	□■□□ □□■	2	□□■□ □□□□
D	□■□□ □■□□	d	□■□□ □□□□	3	□□■□ □□□■
E	□■□□ □□■	e	□■□□ □□■	4	□□■□ □■□□
F	□■□□ □■□□	f	□■□□ □□■	5	□□■□ □■□■
G	□■□□ □■□■	g	□■□□ □■□■	6	□□■□ □■□□
H	□■□□ ■□□□	h	□■□□ ■□□□	7	□□■□ □■□■
I	□■□□ ■□□■	i	□■□□ ■□□■	8	□□■□ ■□□□
J	□■□□ ■□□□	j	□■□□ ■□□□	9	□□■□ ■□□■
K	□■□□ □□■	k	□■□□ ■□□■	.	□□□□ ■□□□
L	□■□□ ■□□□	l	□■□□ ■□□□	,	□□□□ ■□□□
M	□■□□ ■□□■	m	□■□□ ■□□■	:	□□■□ ■□□□
N	□■□□ ■□□□	n	□■□□ ■□□□	;	□□■□ □□■□
O	□■□□ ■□□■	o	□■□□ ■□□■	?	□□■□ ■□□■
V	□■□■ □□□□	p	□■□■ □□□□	!	□□■□ □□□■
Q	□■□■ □□□■	q	□■□■ □□□■	#	□□■□ □□■□
R	□■□■ □□□□	r	□■□■ □□□□	&	□□■□ □□■□
S	□■□■ □□□■	s	□■□■ □□□■	'	□□■□ □□■□
T	□■□■ □■□□	t	□■□■ □■□□	"	□□■□ □□■□
U	□■□■ □□□■	u	□■□■ □□□■	-	□□■□ ■□□■
V	□■□■ □■□□	v	□■□■ □■□□		
W	□■□■ □■□■	w	□■□■ □■□■		
X	□■□■ ■□□□	x	□■□■ ■□□□		
Y	□■□■ ■□□■	y	□■□■ ■□□■		
Z	□■□■ ■□□■	z	□■□■ ■□□■		

2. Controller unit

- The second basic component of the NC system is the controller unit.
- This consists of the electronics and hardware that read and interpret the program of instructions and convert it into mechanical actions of the machine tool.
- The typical elements of a conventional NC controller unit include
 - The tape reader
 - a data buffer
 - Signal output channels to the machine tool
 - Feedback channels from the machine tool
 - The sequence controls to coordinate the overall operation of the foregoing elements.

- The tape reader is an electromechanical device for winding and reading the punched tape containing the program of instructions.
- The data contained on the tape are read into the data buffer
- The purpose of this device is to store the input instructions in logical blocks of information.
- A block of information usually represents one complete step in the sequence of processing elements.
 - For example, one block may be the data required to move the machine table to a certain position and drill a hole at that location.
- The signal output channels are connected to the servomotors and other controls in the machine tool.
- Through these channels, the instructions are *sent to the machine tool from the controller unit*
- To make certain that the instructions have been properly executed by the machine, feedback data are sent back to the controller via the feedback channels.
- The most important function of this return loop is to assure that the table and work-part have been properly located with respect to the tool.
- Sequence controls coordinate the activities of the other elements of the controller unit.
- The tape reader is actuated to read data into the buffer from the tape, signals are sent to and from the machine tool, and so on. - These types of operations must be synchronized and this is the function of the sequence controls.

3. Machine tool

- The third basic component of an NC system is the machine tool or other controlled process.
- It is capable of performing a variety of different operations: drilling, tapping, reaming, milling, and boring
- A variety of machining operations means that a variety of cutting tools are required.
- The tools are kept in a tool drum or other holding device. When the tape calls a particular tool, the drum rotates to position the tool for insertion into the spindle.
- The automatic tool changer then grasps the tool and places it into the spindle chuck.

NC Procedure

To utilize numerical control in manufacturing, the following steps must be accomplished.

1. *Process planning.*

- The engineering drawing of the work-part must be interpreted in terms of the manufacturing processes to be used.
- This step is referred to as process planning and it is concerned with the preparation of a **route sheet**.
- The route sheet is a listing of the sequence of operations which must be performed on the work-part.
- It is called a route sheet because it also *lists the machines through which the part must be routed in order to accomplish the sequence of operations*

2. *Part programming*

- Manual part programming
- Computer-assisted part programming

3. *Tape preparation.*

- A punched tape is prepared from the part programmer's NC process plan

4. *Tape verification*

5. *Production*

Absolute positioning and incremental positioning

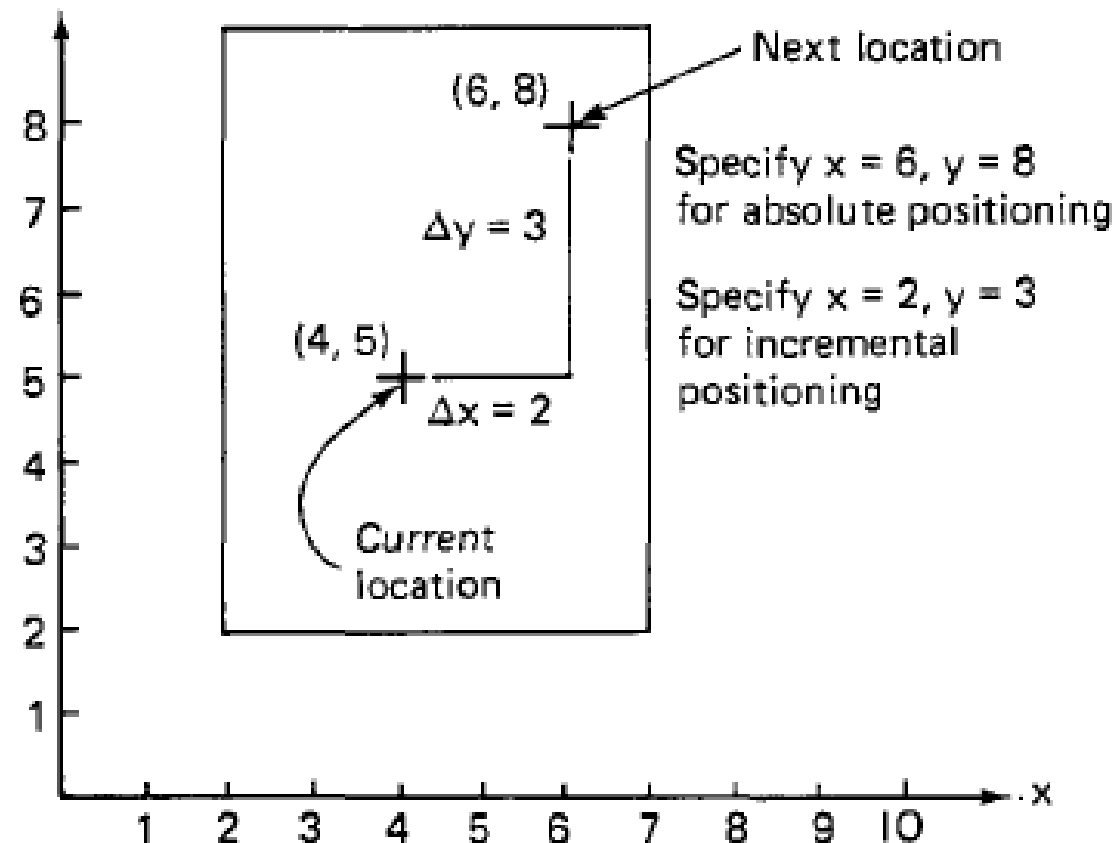


FIGURE 7.6 Absolute versus incremental positioning

NC MOTION CONTROL SYSTEMS

In order to accomplish the machining process, the cutting tool and workpiece must be moved relative to each other

There are three basic types of motion control systems:

Point-to-point NC:

- In PTP, the objective of the machine tool control system is to move the cutting tool to a predefined location
- The speed or path by which this movement is accomplished is not important
- Once the tool reaches the desired location, the machining operation is performed at that position.
- NC drill is a good example of PTP systems

Straight-cut NC

- Straight-cut control systems are capable of moving the cutting tool parallel to one of the major axes at a controlled rate suitable for machining
 - Appropriate for performing milling operations
 - it is not possible to combine movements in more than a single axis direction.
- Therefore, angular cuts on the workpiece would not be possible.
- An example of a-straight-cut operation is shown in Figure

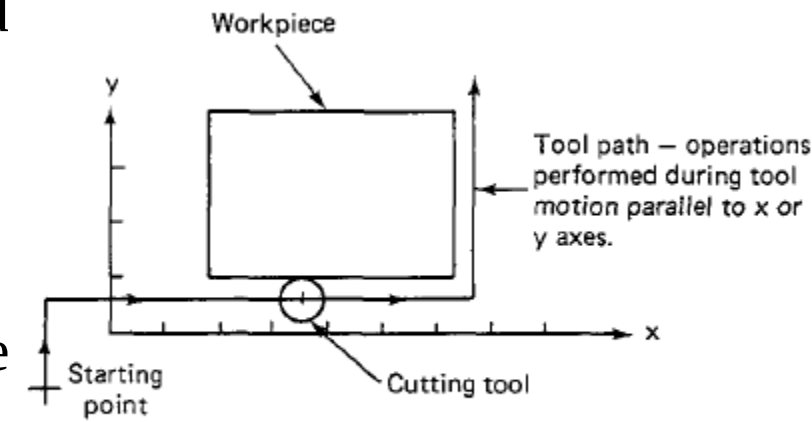


FIGURE 7.8 Straight-cut system.

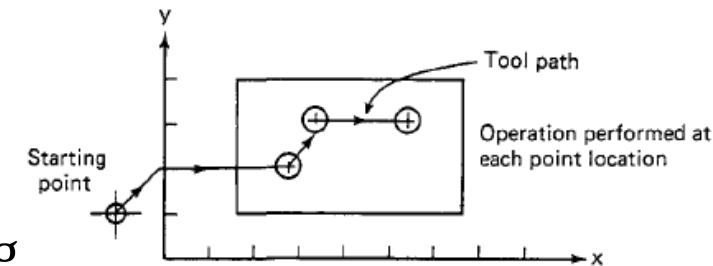


FIGURE 7.7 Point-to-point (positioning) NC system.

NC MOTION CONTROL SYSTEMS

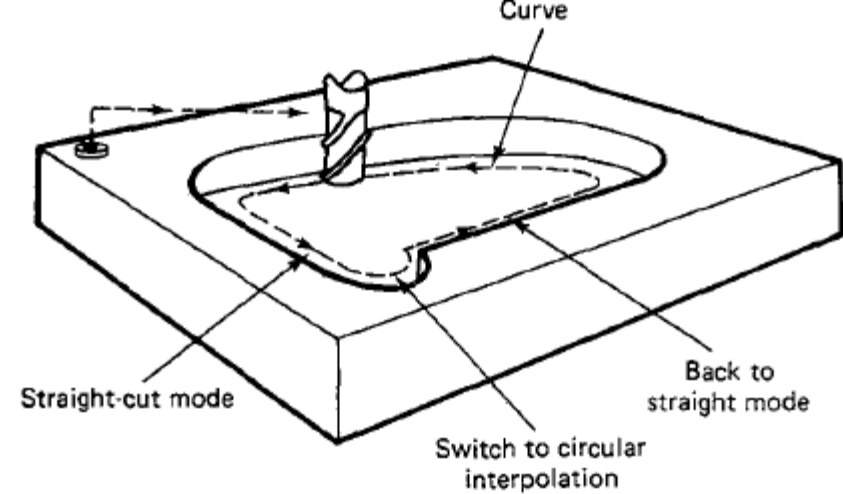


FIGURE 7.9 Contouring (continuous path) NC system for two-dimensional operations. (Reprinted from Olesten [9].)

Contouring NC:

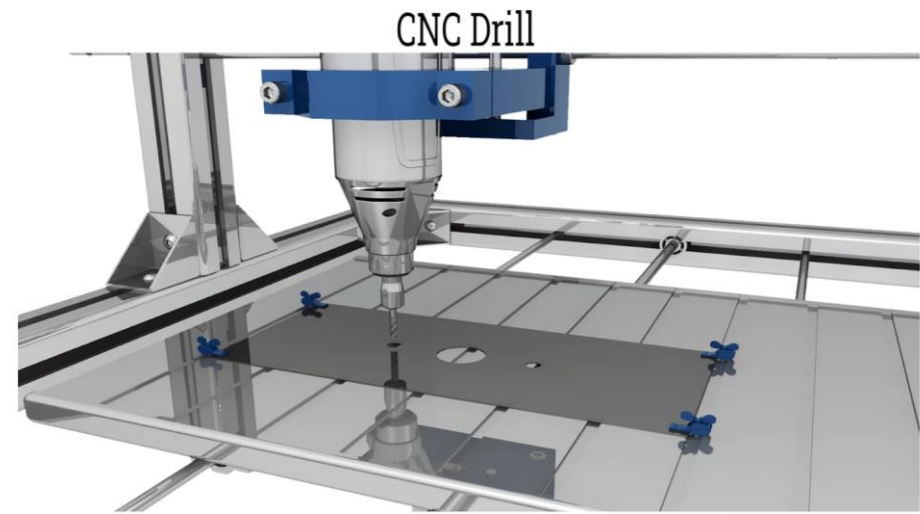
- Contouring is the most complex, the most flexible, and the most expensive type of tool control
- It is capable of performing both PTP and straight-cut operations
- In addition, simultaneous control of more than one axis movement of the machine tool
- The path of the cutter is continuously controlled to generate the desired geometry of the workpiece
- For this reason, contouring systems are also called continuous-path NC systems.
- Straight or plane surfaces at any orientation, circular paths, conical shapes are possible under contouring control

Advantages of NC

- 1. Reduced nonproductive time*
- 2. Reduced fixturing*
- 3. Reduced manufacturing lead time*
- 4. Greater manufacturing flexibility*
- 5. Improved quality control*
- 6. Reduced inventory*
- 7. Reduced floor space requirement*

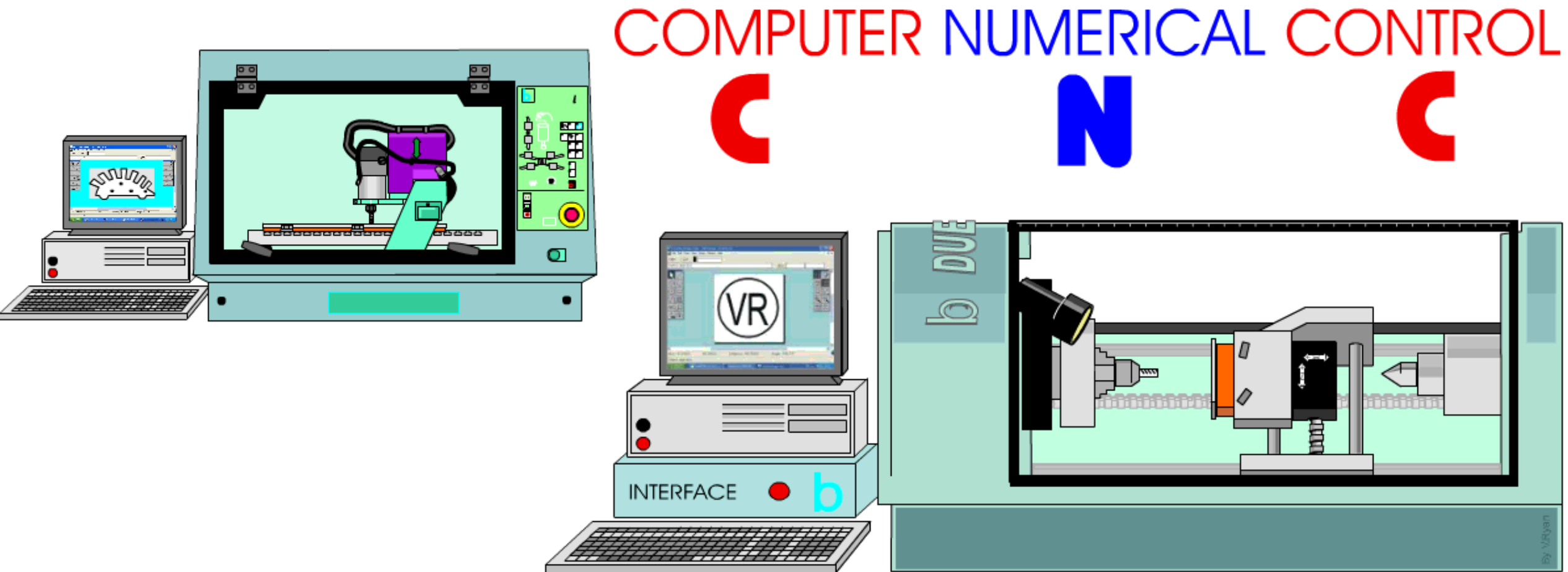
Disadvantages of NC

- 1. Higher investment*
- 2. Higher maintenance cost*
- 3. Finding and/or training NC personnel*



Computer Controls in NC

The computer is being used to refine and improve the NC part programming procedure through interactive graphics and voice programming



PROBLEMS WITH CONVENTIONAL NC

1. *Part programming mistakes.*
2. *Nonoptimal speeds and feeds* - In conventional numerical control, the control system does not provide the opportunity to make changes in speeds and feeds during the cutting process
3. *Punched tape* - *Paper tape is especially fragile, and its susceptibility to wear and tear causes unreliable NC component for repeated use in the shop*
4. *Tape reader* - least reliable hardware component of the machine
5. *Controller.* The conventional NC controller unit is hard-wired, control features cannot be easily altered to incorporate improvements into the unit - Use of a computer as the control device would provide the flexibility to make improvements (becomes software)
6. *Management information.* The conventional NC system is not equipped to provide timely information on operational performance to management

COMPUTER NUMERICAL CONTROL

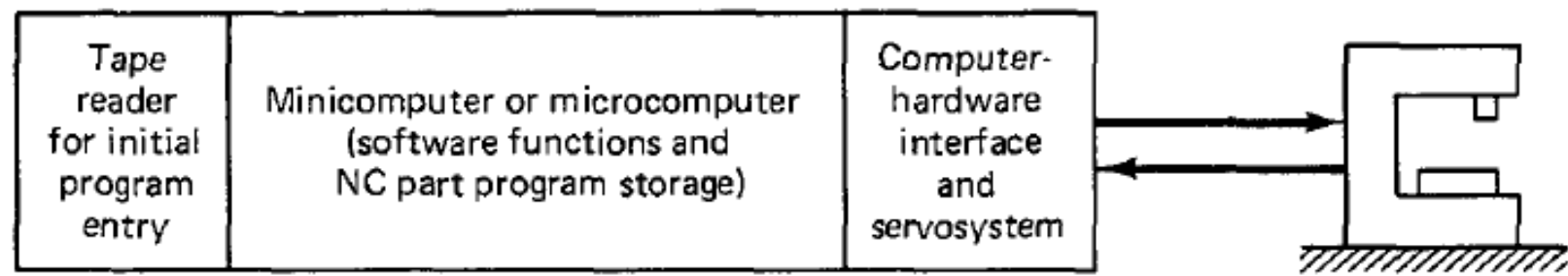


FIGURE 9.1 General configuration of computer numerical control (CNC) system.

- Computer numerical control is an NC system that utilizes a dedicated, stored program computer to perform some or all of the basic numerical control functions.
- Part programs are initially entered in a similar manner.
- Punched tape readers are still the common device to input the part program into the system.
- However, in conventional numerical control, the punched tape is cycled through the reader for every workpiece in the batch.
- With CNC, the program is entered once and then stored in the computer memory.
- Thus the tape reader is used only for the original loading of the part program and data.
- Compared to regular NC, CNC offers additional flexibility and computational capability.
- New system options can be incorporated into the CNC controller simply by reprogramming
- Because of this reprogramming capacity, CNC is often referred to by the term "soft-wired" NC

Functions of CNC

- There are a number of functions which CNC is designed to perform.
- Several of these functions would be either impossible or very difficult to accomplish with conventional NC.

The principal functions of CNC are:

1. Machine tool control
2. In-process compensation
3. Improved programming and operating features
4. Diagnostics

MACHINE TOOL CONTROL

- The primary function of the CNC system is control of the machine tool.
 - This involves conversion of the part program instructions into machine tool motions through the computer interface and servosystem
- The controller consists of the soft-wired controller unit and hard-wired controller unit (logic circuits or data buffer blocks)
 - The capability to conveniently incorporate a variety of control features into the soft-wired controller unit is the main advantage of CNC.
 - Some of the control functions, such as circular interpolation, can be accomplished more efficiently with hard-wired controller unit

Two alternative controller designs in CNC:

1. Hybrid CNC
2. Straight CNC

Hybrid CNC and Straight CNC

Hybrid CNC

- hard-wired controller performs feed rate generation and circular interpolation.
- soft-wired controller performs remaining control functions

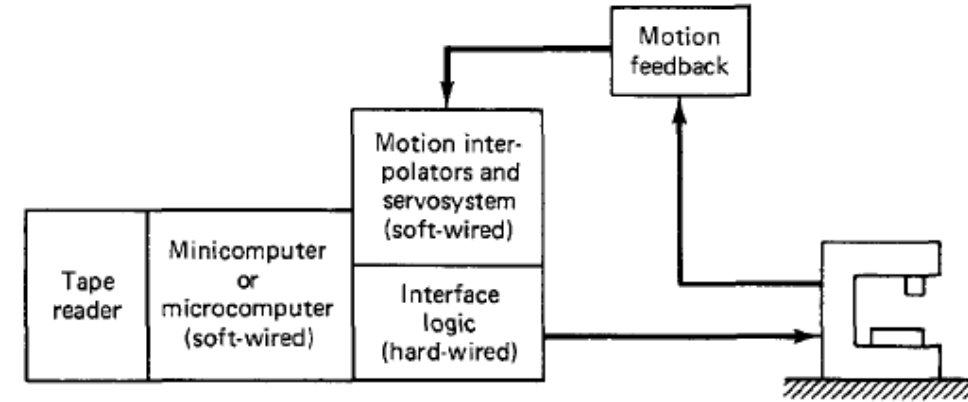


FIGURE 9.2 Hybrid CNC.

Straight CNC

- Computer (soft-wired controller) performs all the NC functions such as Interpolation, tool position feedback
- hard-wired controller required to interface the computer with the machine tool and the operator's console. Interpolation, tool position feedback, and
- the computer required in a straight CNC system must be more powerful than that needed for a hybrid system

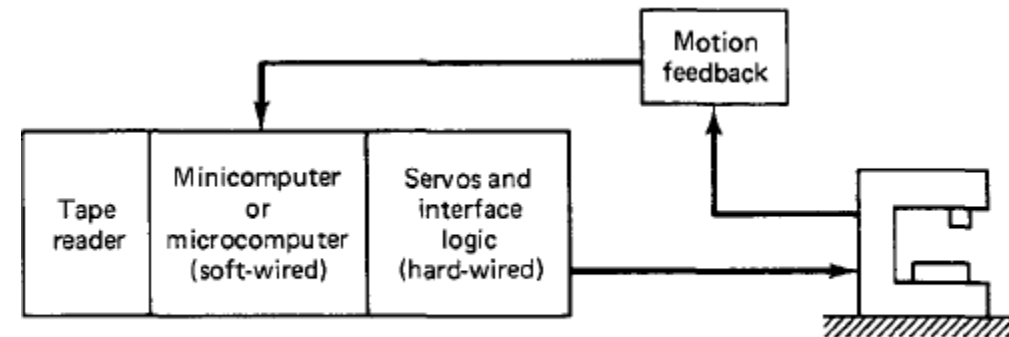


FIGURE 9.3 Straight CNC.

IN-PROCESS COMPENSATION

IN-PROCESS COMPENSATION - dynamic correction of the machine tool motions for changes or errors which occur during processing.

Some of the CNC in-process compensation are:

- Adjustments for errors sensed by in-process inspection
- Re-computation of axis positions when an inspection probe is used to locate a datum reference on a work-part
- Offset adjustments for tool radius and length
- Adaptive control adjustments to speed and/or feed

IMPROVED PROGRAMMING AND OPERATING FEATURES and DIAGNOSTICS

IMPROVED PROGRAMMING AND OPERATING FEATURES - The flexibility of soft-wired control has permitted the introduction of many convenient programming and operating features

- Editing of part programs at the machine.
- Graphic display of the tool path to verify the tape.
- Various types of interpolation: circular, parabolic, and cubic interpolation.
- Support of both U.S. customary units and metric units (International System)

DIAGNOSTICS - NC machine tools are complex and expensive systems

The complexity increases the risk of component failures which lead to system downtime. Therefore avoid downtime as much as possible.

ADVANTAGES OF CNC

- 1. The part program tape and tape reader are used only once*
- 2. Tape editing at the machine site*
- 3. Metric conversion*
- 4. Greater flexibility*
- 5. User-written programs*
- 6. Total manufacturing system*